Exception Investigation Report – Air Arabia Alphanumeric Flight Number Handling

Story Summary

Objective: Investigate and resolve an application exception occurring during flight search rule processing for Air Arabia flights.

Exception Type: Rule Engine Exception

Supplier: Air Arabia

Credential ID: 280

Occurrence Date: 31-May-2026

Affected Component: Rule Engine / Flight Option Processing

Route: JED → HBE (30-06-2026)

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1. Root Cause Investigation

Observed Exception

```
RuleEngineException Oneway Supplier: AirArbia, Credential Id: 280
Exception Message: For input string: "621D"
```

Stack Trace Entry Point

```
RuleSimulationHelper.ruleApplyFlightOption()
RuleSimulationHelper.processOdeysysFlightOption()
```

Execution Flow

```
Flight Search Response
    ↓
processOdeysysFlightOption()
    ↓
ruleApplyFlightOption()
    ↓
Flight Number Processing
    ↓
Integer Parsing
    ↓
NumberFormatException
```

Root Cause

The Rule Engine currently assumes that all supplier flight numbers are numeric values and therefore converts flight numbers into Integer values during rule processing.

Air Arabia returned a flight segment containing an alphanumeric flight number:

```
"flightNumber": "771D"
```

When the application attempted to convert this value into an Integer, a `NumberFormatException` occurred, causing the Rule Engine execution to fail.

Example:

```
Integer.parseInt("771D");
```

Result:

```
NumberFormatException: For input string: "771D"
```

Technical Finding

The assumption that all suppliers provide numeric flight numbers is invalid.

Air Arabia legitimately returns alphanumeric flight numbers, which are supported by airline industry standards.

2. Impact Assessment

Affected Functionality

- Flight Search
- Rule Engine Processing
- Flight Option Evaluation
- Rule Simulation

User Impact

Affected Air Arabia itineraries may:

- Fail during rule evaluation.
- Not receive applicable business rules.
- Be excluded from search results.
- Produce unexpected search behavior.

Operational Impact

- Error logs generated in production.
- Increased monitoring noise.
- Reduced reliability of rule processing for affected suppliers.

Frequency

The issue occurs whenever:

- Air Arabia returns a flight number containing alphabetic characters.
- Any future supplier returns alphanumeric flight numbers.

The issue is therefore supplier-data dependent and potentially affects multiple suppliers.

3. Solution Analysis

Option 1 – Support Flight Numbers as String (Recommended)

Description

Modify the application to treat flight numbers as String values throughout rule processing instead of Integer values.

Required Changes

Domain Model

Change flight number datatype:

```
Integer flightNo
```

to:

```
String flightNo
```

Rule Engine

Update rule conditions to compare string values.

Example:

Current:

```
flightNo == 771
```

Updated:

```
flightNo == "771D"
```

or

```
flightNo in ("771D", "771")
```

Advantages

- Supports all airline flight number formats.
- Aligns with supplier responses.
- Minimal architectural changes.
- Future-proof for additional suppliers.

Disadvantages

- Existing Drools rules must use String values.
- Rule authors must quote flight numbers.
- Existing numeric flight-number rules may require review.

Option 2 – Introduce Flight Number Reference IDs via CRUD

Description

Create a master-data entity for flight numbers.

Each flight number will have an internal numeric identifier.

Example:

ID	Flight Number
1	771D
2	146
3	621D

Rules will reference the internal ID rather than the supplier flight number.

Advantages

- Rules continue using Integer values.
- Centralized management.
- Easier future administration through UI.

Disadvantages

- Requires new database tables.
- Requires CRUD screens.
- Additional maintenance effort.
- Increased implementation complexity.
- Requires operational management of flight-number records.

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4. Proposed Solution

Recommended Approach

Option 1 – String-Based Flight Number Handling

Reasoning:

- Flight numbers are identifiers rather than mathematical values.
- Airline industry standards allow alphanumeric flight numbers.
- Lowest implementation effort.
- Eliminates similar future issues from other suppliers.

The system should be updated to treat flight numbers as String values throughout Rule Engine processing.